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Benjamin Schmidt

20 August 1988

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Skills

- *Big Data* Management and Analysis
- Data Visualization
- Full Stack Development
- Server Administration
- High-Performance Computing
- Scientific Writing
- Communication and Teamwork

Education

2025 Ph.D. – TU Berlin (Magna c.l.)

2019 M.Sc. – Humboldt University Berlin

2016 B.Sc. – CAU Kiel / GEOMAR

2008 Abitur – Waldorf School (Flensburg)

Languages

German
C2

English
C2

Spanish
B2

French
B1

Language proficiency according to CEFR.

EXPERIENCE

02/24 - today **Chief Scientific Officer (CSO)** Alganize GmbH

Mandate: Build scientific and analytical strategy for climate-intelligence offerings in the BioTech context.
Responsibilities: Advise leadership on data-product direction, measurable analysis goals, and evidence standards for external communication.
Translate climate and station time-series outputs into decision-ready narratives for non-technical stakeholders and partners.

10/19 - today **Lecturer** FU Berlin - TU Berlin

Courses: Statistics 101, Statistics with R, Climatology, Urban Atmosphere, Scientific Programming, Remote Sensing Data Analysis.

08/19 - 02/26 **Research Associate (PhD)** Technische Universität Berlin - Climatology

Initiative: MaRESS - Research mapping and analytics platform for Earth-system data
Responsibilities: Core contributor to an analytical and visualization platform for spatial and graph data, enabling teams to identify high-value datasets and research connections faster.
Led end-to-end full-stack delivery from data model and API design to interactive map and network interfaces, turning complex scientific structures into clear decision support for diverse stakeholders.
Built NLP-based metadata enrichment workflows that improved discoverability and reusability, a pattern directly transferable to enterprise knowledge graphs and document intelligence use cases.

Initiative: DARWIN – Climate-risk intelligence for water resources in the Galápagos.
Responsibilities: Managed and analyzed large atmospheric model datasets (NetCDF) to produce decision-ready climate insights under uncertainty.
Designed and operated end-to-end data pipelines from ingestion to visualization, improving reproducibility and reducing manual cycle time for recurring analyses.
Developed data-intensive applications and dashboards with Plotly/Dash and Panel for exploratory analysis, scenario comparison, and cross-functional communication.
Expanded high-performance computing workflows for large-scale model execution and post-processing, including reliable job orchestration, storage management, and throughput optimization.
Published peer-reviewed results and translated technical findings into practical recommendations relevant to generic business risk, resilience, and planning contexts.

Initiative: $[UC]^2$ - Urban Climate Under Change
Responsibilities: Data analysis and data management for complex, heterogeneous urban-climate datasets.
Full stack development of web platforms for management and exchange of large datasets with Django and Nuxt.
Development of the data visualization website UCO Berlin.
Planning and development of visualization and analysis tools and the $[UC]^2$ data standard for NetCDF atmospheric data, balancing interoperability and real-world workflow constraints.

Initiative: Q-TiP - Quaternary Tipping Points of Lake Systems in the Arid Zone of Central Asia.
Responsibilities: Modeling, analysis and visualization of climatological data using WRF model and Python.

02/19 - 07/19 **Research Assistant** Potsdam Institute for Climate Impact Research PIK - ISIMIP

Initiative: ISI-CFACT - Counterfactual climate data synthesis from historical observations
Responsibilities: ATTRICI design and application of algorithms for generating counterfactual climate datasets ("what-if" baselines) from large-scale historical data in Python.

02/17 - 01/19 **Student Research Assistant** Humboldt-Universität zu Berlin - Climate Geography

Initiative: Heat waves in Berlin – Climate projections.
Responsibilities: Application of the mesoscale climate model COSMO-CLM with the urban canopy scheme *Double Canyon Energy Parameterization*; data collection, processing, and analysis.